

# AI-SUPPORTED WRITING TASK

## Write – Revise – Reflect

The aim of this task is to improve your English by writing, revising your text, working with AI feedback and reflecting on your learning.

### YOUR TASK AT A GLANCE

- Write a text of 150–200 words.
- Produce three versions: V1, V2 and V3.
- Use AI twice to receive feedback on your language.
- You are responsible for your text. Do not simply copy AI suggestions if you do not understand them.
- Finish with a short language reflection.

You may use an English dictionary. *→ like Oxford Dictionary or Cambridge Dictionary } provide relevant details + example sentences*  
**Do not use translation tools.** Do not ask AI to write or rewrite your text for you.

## STEP 1: VERSION 1

Choose one topic from the list below. You may also choose another topic, but make sure that you can write about it in 150–200 words.

- My degree programme: What do I like about it? Why did I choose it?
- A hobby or activity I really enjoy
- A place that is important to me
- A person who has influenced me
- Something I am good at or something I would like to learn
- A food or dish I really like - including how to prepare it
- A small topic related to society, nature, university or everyday life that interests me

### A SIMPLE STRUCTURE FOR YOUR TEXT

Introduction: What is your topic?

Main part: Give information, reasons, examples or details.

Ending: Why is this topic interesting or important (to you and others) ?

**My topic:** Service-Interface-Identification during the Migration of Monoliths to Microservices

*(It is the topic of my seminar work which I am going to write in the upcoming semester.)*

**Plan / key words (optional):** Software Migration, Monoliths, Microservices, Service Identification and Packaging, Service-Interface-Identification

**Plan: Introduce the reader my seminar work**

**1. Introduction and Context**

*"I want to introduce you to the topic of seminar work..."*

**2. Explain briefly Monoliths and Microservices**

**3. Talk about Migration and it's phases**

**4. Explain focus of the seminar work**

*"The topic of my seminar work will be included in the Packaging phase..."*

**5. Explain the goal of the seminar work**

*"The goal of my seminar work is to go through current literature to examine the current state of research in regarding to Service-Identification-Identification."*

**6. Explain applied methodology**

*"The seminar work will apply the well-established SLR-Methodology proposed by Kitchenham."*

*"The seminar work will provide an overview and roadmap of the research area of Service-Interface-Identification."*

**Note during/after writing my first version:**

150 to 200 words are too less to write about all aspects I wanted to include. I focused on only to explain what my seminar work is about trying to stay short, precise and understandable. I left out step 5 and 6 from my above-mentioned plan.

## VERSION 1 (150–200 words)

I want to introduce you briefly to the topic of my seminar work: “Service-Interface-Identification during the Migration of Monoliths to Microservices”. I find this topic interesting and relevant as software is evolving on the one hand in a quite fast manner, but organisations struggle on the other hand to migrate their software as it is a resource intensive and complex task (Trabelsi et al., 2025).

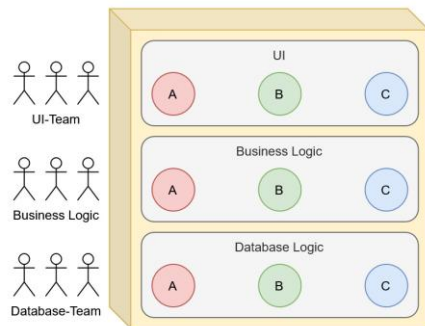


Figure 1: Illustration of a monolith. Original illustration by (Nockemann, 2025).

*Microservices* seeks to overcome the disadvantages of monoliths and have therefore gained significant attraction (Abgaz et al., 2023; Gysel et al., 2016). They usually decompose the software around business capabilities which can be developed and deployed independently which is illustrated in Figure 2 (Lewis & Fowler, 2014).

During the migration process the identification of possible microservice candidates is extensively studied whereas other phases like *Microservice Packaging* are less extensively studied. This phase seeks to encapsulate identified candidates into functional microservices. (Trabelsi et al., 2025)

My seminar work seeks to enlighten exactly this area of research by focusing on the identification of the interfaces of the provided microservice candidates.

*Monoliths* which are currently the predominant used software architecture (Abgaz et al., 2023) decompose the software and developer teams around technical layers which is illustrated in Figure 1 (Lewis & Fowler, 2014; Nockemann, 2025).

When the monolith becomes larger and more complex it encounters several challenges in *adaptability*, *flexibility*, *scalability* and *deployability*. Whenever the software has to be slightly adapted, all layers may be involved hindering the above mentioned desired features. (Abgaz et al., 2023)

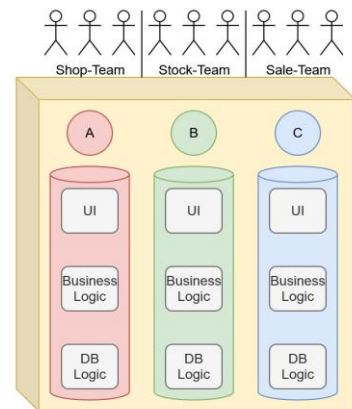


Figure 2: Illustration of microservices. Original illustration by (Nockemann, 2025)

## References:

- Abgaz, Y., McCarren, A., Elger, P., Solan, D., Lapuz, N., Bivol, M., Jackson, G., Yilmaz, M., Buckley, J., & Clarke, P. (2023). Decomposition of Monolith Applications Into Microservices Architectures: A Systematic Review. *IEEE Transactions on Software Engineering*, 49(8), 4213–4242. <https://doi.org/10.1109/TSE.2023.3287297>
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- Trabelsi, I., Mahmoudi, B., Minani, J. B., Moha, N., & Guéhéneuc, Y.-G. (2025). A Systematic Literature Review of Machine Learning Approaches for Migrating Monolithic Systems to Microservices. *IEEE Transactions on Software Engineering*, 51(11), 2972–2995. <https://doi.org/10.1109/TSE.2025.3603897>

## STEP 2: GET AI FEEDBACK

Use the AI tool you normally work with. Copy and paste the prompt below. Then paste the complete feedback into this document.

### PROMPT 1 - COPY AND PASTE

Please analyse my text as an English teacher. I am an English learner, so please adapt your feedback to my language level. Do not rewrite or correct my text. Identify my language mistakes and explain briefly what is wrong. Then identify 1–2 general patterns in my mistakes, for example word order, verb forms, articles, prepositions, negation or vocabulary. Do not give me a corrected version of my text.

**Which AI did I use?** Google Gemini

**AI feedback – copy and paste the complete feedback here:**

Here is an analysis of your text to help you improve your English.

### Identified Language Mistakes

1. **"Monoliths which are currently the predominant used software architecture..."**
  - **Explanation:** *"Predominant used"* is incorrect word choice/form. You need an adverb like **"predominantly"** to describe how it is used, or an adjective like **"predominant"** on its own (e.g., *"the predominant software architecture"*).
2. **"...decompose the software and developer teams around technical layers which is illustrated in Figure 1..."**
  - **Explanation:** *"Monoliths"* is plural, but *"which is illustrated"* uses a singular verb (*is*). Because it refers to the general concept or multiple monoliths, the verb agreement needs attention. Additionally, missing a comma before *"which"* makes the relative clause harder to read.
3. **"...it encounters several challenges in adaptability, flexibility, scalability and deployability."**
  - **Explanation:** Spelled *"deployability"* incorrectly (typo: should be *deployability*).
4. **"Microservices seeks to overcome..."**
  - **Explanation:** *"Microservices"* is a plural noun, so it needs a plural verb (**"seek"** instead of *"seeks"*).
5. **"...and have therefore gained significant attraction..."**
  - **Explanation:** Incorrect word choice. The standard phrase in English is **"gained traction"** or **"gained popularity"** when something becomes widely accepted or used.
6. **"...which can be developed and deployed independetly..."**
  - **Explanation:** Spelled *"independetly"* incorrectly (typo: missing an 'n', *independently*).
7. **"My seminar work seeks to enlight exactly this area..."**

- **Explanation:** Incorrect verb form/word choice. The correct verb is "**enlighten**" (or more commonly in academic writing, "*shed light on*" or "*explore*").

## General Patterns in Your Mistakes

### 1. Subject-Verb Agreement (Plural vs. Singular)

- You tend to pair plural subjects with singular verbs. For example, writing "*Microservices seeks*" instead of "*Microservices seek*". When writing, always check if your subject ends in an **-s** (plural) and ensure the verb matches it properly.

### 2. Word Choice & Collocations (Vocabulary)

- Your writing shows a high level of vocabulary, but some word combinations are slightly off. Phrases like "*gained attraction*" (instead of *gained traction*) or "*enlight*" (instead of *enlighten/shed light on*) are near-matches that sound a bit unnatural to a native speaker. Paying extra attention to fixed academic expressions will quickly solve this!

Would you like to practice fixing any of these specific areas with a short exercise?

## Think about the feedback

What do I understand?

I understand most of the mistakes the AI pointed out. I don't understand the explanation of the second mistake.

Which mistakes do I want to correct?

I want to correct all spelling and word selection mistakes.

Where do I want to rewrite?

I want to rewrite the second paragraph a bit to make it more clear.

What is unclear?

The explanation of the second mistake is unclear for me. The mentioned relative sentence "which is illustrated in Figure 1." doesn't correspond for me to "Monoliths" but rather to the decomposition into technical layers.

## STEP 3: VERSION 2

Now improve your text yourself. Use the AI feedback, but make your own decisions. Do not copy suggestions that you do not understand.

I want to introduce you briefly to the topic of my seminar work: “Service-Interface-Identification during the Migration of Monoliths to Microservices”. I find this topic interesting and relevant because software is evolving on the one hand in a quite fast manner, but on the other hand organisations struggle to migrate their software as it is a resource intensive and complex task (Trabelsi et al., 2025).

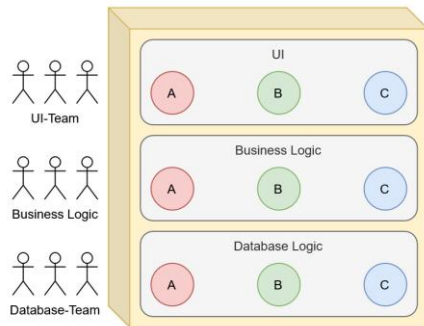


Figure 1: Illustration of a monolith. Original illustration by (Nockemann, 2025).

*Microservices* seek to overcome the disadvantages of monoliths and have therefore gained significant traction (Abgaz et al., 2023; Gysel et al., 2016). They usually decompose the software around business capabilities which can be developed and deployed independently which is illustrated in Figure 2 (Lewis & Fowler, 2014).

During the migration process the identification of possible microservice candidates is extensively studied whereas other phases like *Microservice Packaging* are poorly studied. This phase seeks to encapsulate identified candidates into functional microservices. (Trabelsi et al., 2025)

My seminar work seeks to explore this area of research by focusing on the identification of the interfaces of provided microservice candidates.

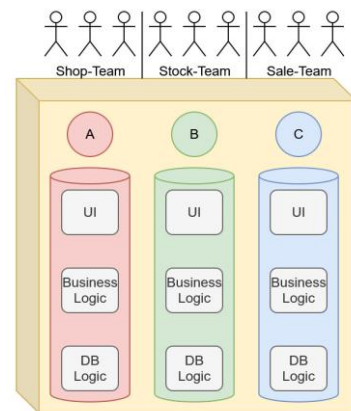


Figure 1: Illustration of microservices. Original illustration by (Nockemann, 2025)

### References:

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## STEP 4: GET AI FEEDBACK AGAIN

### PROMPT 2 – COPY AND PASTE

Please analyse this revised version. Do not rewrite it. Tell me which problems from my first version I have successfully corrected and which language problems still occur. Identify one or two language areas I should continue to work on.

Which AI did I use? Google Gemini

AI feedback – copy and paste the complete feedback here:

### Successfully Corrected Issues

- **Adverb Choice:** You corrected "*predominant used*" to "**predominantly used**".
- **Sentence Structure & Punctuation:** You restructured the paragraph on monoliths, removing the awkward relative clause ("*which is illustrated in Figure 1*") and correctly placing "*As illustrated in Figure 1*" at the beginning of the sentence.
- **Spelling:** Both typos ("*deployablitiy*" and "*independetly*") were fixed to "**deployability**" and "**independently**".
- **Subject-Verb Agreement:** You changed "*Microservices seeks*" to the correct plural form "**Microservices seek**".
- **Collocations / Natural Phrasing:**
  - You updated "*gained attraction*" to the idiomatic expression "**gained traction**".
  - You replaced "*enlight*" with "**explore**", which fits formal academic context well.

### Remaining Language Problems

1. "**Whenever the software has to be slightly adapted, all layers may be involved hindering the above mentioned desired features.**"
  - **Problem:** Punctuation / Clause structure. Adding a participle clause ("*hindering...*") directly onto the sentence creates a slight run-on feeling. A comma is needed before "**hindering**" to properly join this participle clause to the main clause.
2. "**...which can be developed and deployed independently which is illustrated in Figure 2...**"
  - **Problem:** Repetitive relative clauses and missing punctuation. Using "*which*" twice in close succession ("*...which can... which is...*") sounds repetitive. Additionally, a non-defining relative clause like "*, which is illustrated in Figure 2*" requires a comma before "**which**".
3. "**...focusing on the identification of the interfaces of provided microservice candidates.**"
  - **Problem:** Missing article. You need the definite article "**the**" before "*provided microservice candidates*" because you are referring to specific candidates that were identified earlier.

## Key Areas to Work On

### 1. Punctuation with Relative and Participle Clauses

- Pay attention to commas when adding extra information to a sentence. Non-defining relative clauses starting with "**which**" and active participle clauses starting with an "**-ing**" verb (like "*hindering...*") almost always require a comma preceding them to separate them clearly from the main clause.

### 2. Definite Articles ("*the*") with Specific Nouns

- Practice identifying when a noun phrase refers to a specific, previously mentioned object or group (e.g., "*the provided candidates*") rather than a general concept.

Would you like to focus on clause punctuation or article usage for a quick follow-up exercise?

## My response to the second feedback

Which problems did I successfully correct?

I successfully corrected all mistakes, but I invented new ones and the AI mentioned mistakes which it hasn't mentioned before.

Which problems still occur?

Problems with relative clauses.

Which language area do I want to continue working on?

I want to work on practising using the Definite Article. Even without the AI feedback, I noticed that I still have problems to decide whether "the" is appropriate or not. I also struggle to distinguish the meaning when I use "the" or when I left it out.

## STEP 5: VERSION 3 – FINAL VERSION

If you find the second AI feedback useful, use it to improve your text once more. Version 3 is your final version.

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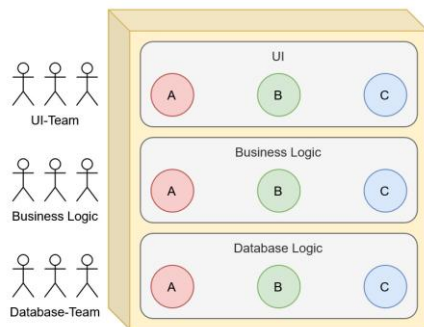


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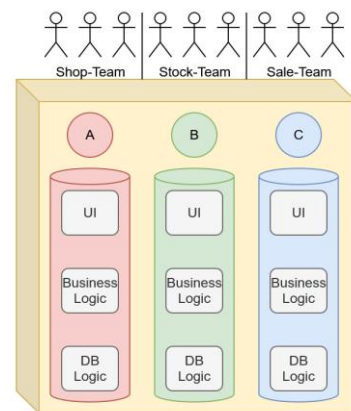


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### References:

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## STEP 6: MY LANGUAGE REFLECTION

Answer the questions honestly and specifically. Short answers are fine. The aim is to show what you have learned about your English.

### 1. One mistake I often make is ...

that I tend to stack relative clauses starting with “which” making sentences unnecessary complex and difficult to read.

Another thing is that I tend to wrongly remember and use English specific formulations like “gained traction” which I wrongly remember as “gained attraction”.

### 2. I learned that ...

I should keep my sentences shorter and more precise instead of stacking ideas. Sometimes I want include several ideas, details and pieces of information in just one single sentence. I learned to think about which pieces of information I want the sentence to convey, instead of trying to just cluster details and pieces of information. I learned to organise more my sentence structure, combining suitable pieces of information in one sentence and splitting up sentences.

### 3. One thing I corrected myself was ...

that I split up overly complex sentences, so that one sentence focus to convey one single aspect rather than clustering several different aspects and ideas.

### 4. I still have problems with ...

using the Definite Article “the” correctly and to distinguish the difference in meaning when I use it or when I leave it out.

### 5. In the future, I want to practise ...

the usage of the Definite Article and I want to learn more special English related terms like “gained traction”.

### 6. One useful thing I learned from the AI feedback was ...

that it can help me improving my writing language when properly prompted. I wish to use it also to show me how I can improve my sentence structure.

I know that the prompt told the AI to just only focus on language related problems, but I would also use the AI to suggest higher-level improvements going beyond grammar.

### 7. One AI suggestion I did not use was ... because ...

I didn't use the suggestion to write “which are illustrated in Figure 1.” As it sounds odd for me. The actual solution was to split up and reformulate the sentence avoiding the stacking of “which”-clauses.

### FINAL CHECK

- ☒ V1, V2 and V3 are included.
- ☒ The two complete AI feedbacks are included.
- ☒ I made my own decisions about the AI suggestions.
- ☒ My final text is 150–200 words (approximately). *without citations and figure slightly more: around 215*
- ☒ I completed the reflection.
- ☒ I understand the language changes I made.

**25 points**

Progress V1 → V3: 7 points | Final version: 7 points | AI-supported learning & reflection: 11 points